

Network Forensics Guide for Thaghrah

Description

Network Forensics is the process of capturing, recording, and analyzing network traffic to investigate security incidents, troubleshoot problems, or reconstruct events. It involves examining packet captures (PCAPs) to understand what happened on the network — such as detecting intrusions, tracing data exfiltration, identifying malicious behavior, or recovering lost information.

Unlike the other protocol-specific categories, **Network Forensics** challenges combine multiple protocols and require broader investigative thinking. Analysts look at the "big picture" across different layers of the network stack to piece together timelines, user actions, and attacker techniques.

Key Components of Network Forensics

1. Core Objectives

- Reconstruct events and timelines from network traffic
- Identify sources and destinations of communication
- Detect anomalies, suspicious patterns, or data leaks
- Recover transferred files, credentials, or sensitive information
- Correlate activity across multiple protocols

2. Common Techniques

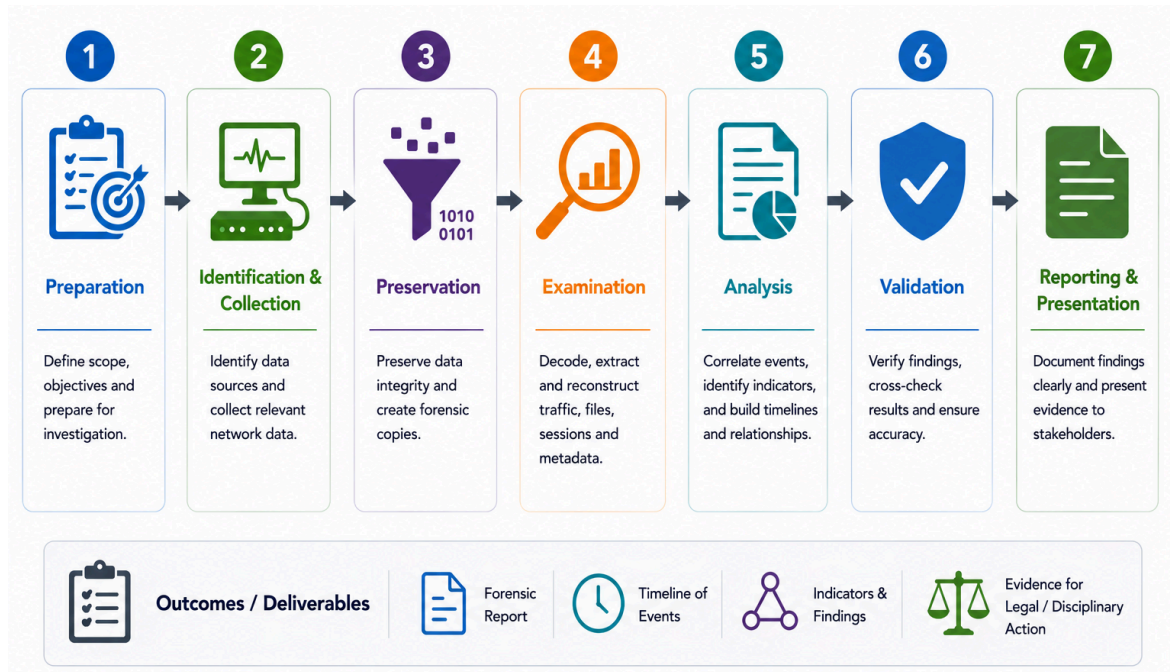
- **Traffic Analysis:** Understanding normal vs abnormal behavior
- **Protocol Analysis:** Examining HTTP, DNS, TCP, FTP, SMTP, TLS, etc.
- **Session Reconstruction:** Reassembling conversations (e.g., following TCP streams)
- **Artifact Recovery:** Extracting files, images, or payloads from captures
- **Timeline Building:** Ordering events based on timestamps

3. Important Concepts

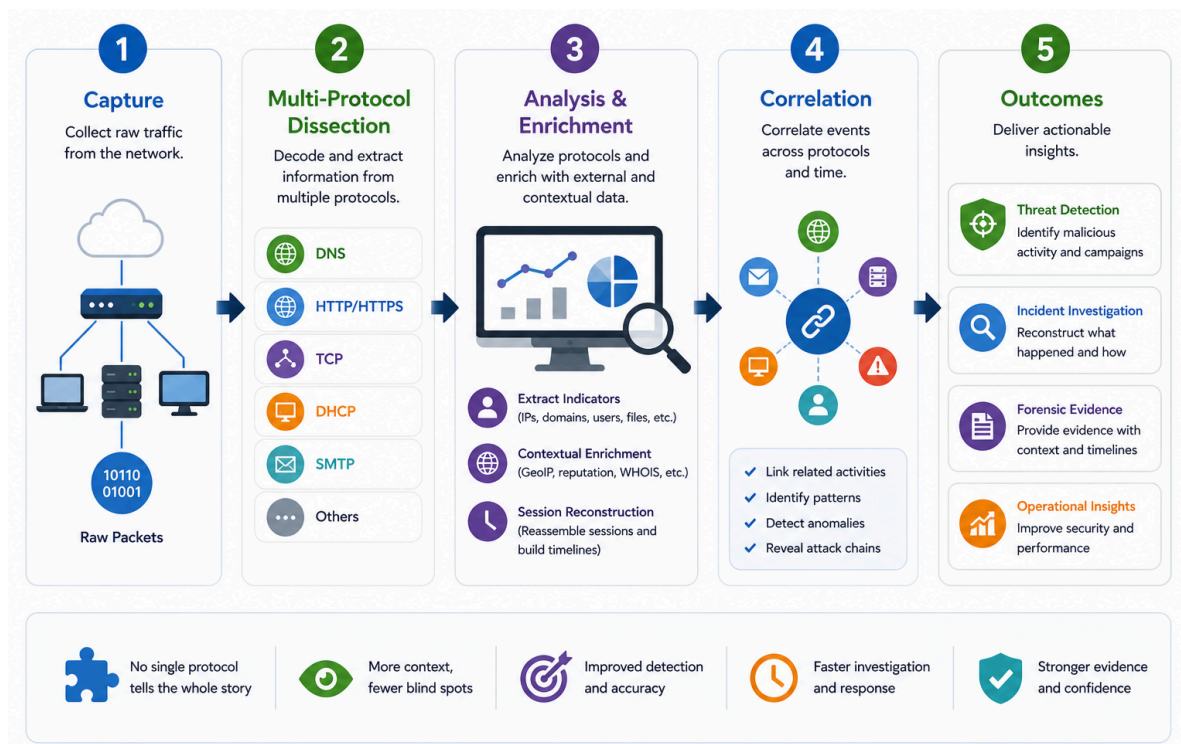
- **Indicators of Compromise (IoCs):** Suspicious IPs, domains, unusual ports, large data transfers
- **Data Exfiltration:** Unauthorized transfer of sensitive data out of the network
- **Multi-Protocol Correlation:** Connecting clues from DNS queries → HTTP requests → FTP uploads, etc.
- **Encryption Challenges:** Analyzing traffic before/after TLS or identifying encrypted tunnels

Visualizations

Network Forensics Investigation Workflow:



Multi-Protocol Analysis Overview:



What Is Network Forensics?

<https://youtu.be/q0X3ZBDxIAs?si=MbgXs5dwL2qJsKk2>